Kim (2023), *Personal Networks, State Financial Backing and Foreign Direct Investment* — SH Day 2: Regression Extensions

Machine-Learning Workshop teaching example

2026-07-21

1 Study and source

Citation. Seungjun Kim. "Personal Networks, State Financial Backing and Foreign Direct Investment." *Comparative Political Studies* 56(7), 2023.

Replication source. Harvard Dataverse (Comparative Political Studies collection), DOI `10.7910/DVN/GFNDOS` (https://doi.org/10.7910/DVN/GFNDOS). The dataset is a cross-section of South Korean listed firms; the posted replication `.R` script estimates the main results with plain `lm()` OLS.

Main model. Table 1 of the paper reports OLS regressions explaining the *amount of a firm's outbound FDI projects* (logged). The theoretical claim: firms whose boards have ties to state-owned banks obtain financial backing that translates into more foreign investment. The fully-specified column (Table 1, model 4) is:

$$\begin{aligned}\texttt{FDIProjects\_Amount} \sim\ & \texttt{Board\_Ties\_to\_State\_Banks} + \texttt{Firm\_Size} + \texttt{Debt\_Ratio} \\ & + \texttt{Profitability} + \texttt{Chaebol} + \texttt{factor(sic\_category)}.\end{aligned}$$

Outcome: continuous (logged FDI project amount). Model type: **OLS**. Estimation N = 703 (after listwise deletion), across nine one-digit SIC industry categories.

What this Day-2 tutorial does. We keep the exact reproduction of Table 1, then treat the *same outcome* with the full toolkit of regression extensions: multiple-regression interpretation, the complete residual-diagnostic suite, a logistic-regression / MLE aside, and a sequence of increasingly flexible mean functions in one key predictor (polynomial → spline → GAM), all compared honestly out of sample. This study is an ideal vehicle because firm size has a strongly **nonlinear** relationship to FDI that a straight-line OLS term cannot capture.

2 Descriptive analysis

A thorough exploratory pass earns its keep here: the outcome's extreme shape and the firm-size predictor's curvature are exactly what motivate every extension on Day 2, so we want to *see* them before we model them. We proceed in six movements — (1) the dataset and a variable dictionary; (2) the outcome in depth; (3) univariate predictor distributions; (4) missing-data analysis; (5) bivariate relationships with the outcome; and (6) multivariate structure and collinearity — closing with a narrative of what the descriptives imply for the modeling.

2.1 Dataset, provenance, and variable dictionary

Unit of analysis. The row is a **South Korean publicly-listed firm**. The design is a single 2011 cross-section: the predictors are measured at (or as of) 2011, and the outcome accumulates each firm's outbound foreign-direct-investment activity through 2016. There is no panel/time dimension — every firm contributes one observation.

Provenance and sampling. Firms are the KOSPI/KOSDAQ-listed universe used in Kim (2023), assembled from Worldscope financials, board-membership records used to code personal ties to state-owned banks, and FDI project filings aggregated to 2016. The replication file is the posted Stata `.dta` from Harvard Dataverse (DOI `10.7910/DVN/GFNDOS`); the authors' own `.R` script fits the main table with plain `lm()`. There is no probability sampling — this is a near-census of listed firms — so the relevant inferential frame is the population of listed Korean firms, not a random draw from a larger pool.

Size and shape. The raw file holds 732 firms and 17 columns; the modeling frame `md` (the six Table-1 predictors plus the outcome, listwise complete) has $N = 703$ firms and 7 columns. Listwise deletion drops 29 firms (Section on missingness below). The dictionary lists the outcome and the six modeling predictors.

Table 1: Variable dictionary for the outcome and Table-1 modeling predictors.

Variable	Role	Type	Description
FDIProjects_Amount	Outcome	continuous	Logged cumulative outbound FDI project amount, 2016
Board_Ties_to_State_Banks	Treatment	binary	1 if a board member is tied to a state-owned bank
Firm_Size	Predictor	continuous	Scaled 2011 market capitalisation (firm size)
Debt_Ratio	Control	continuous	2011 debt-to-assets ratio
Profitability	Control	continuous	2011 return on equity (ROE, percent)
Chaebol	Control	binary	1 if firm belongs to a chaebol conglomerate
sic_category	Control	factor (9)	One-digit SIC industry code

2.2 The outcome in depth

The outcome is **logged outbound FDI project amount**. It is continuous in principle but massively **zero-inflated**: 81% of firms have exactly zero (they never invested abroad), so the distribution is a spike at zero next to a right-skewed continuum of positive values that runs out to about 10. The left panel below shows the full distribution (histogram with a density overlay); the right panel shows only the positive mass, which is itself right-skewed.

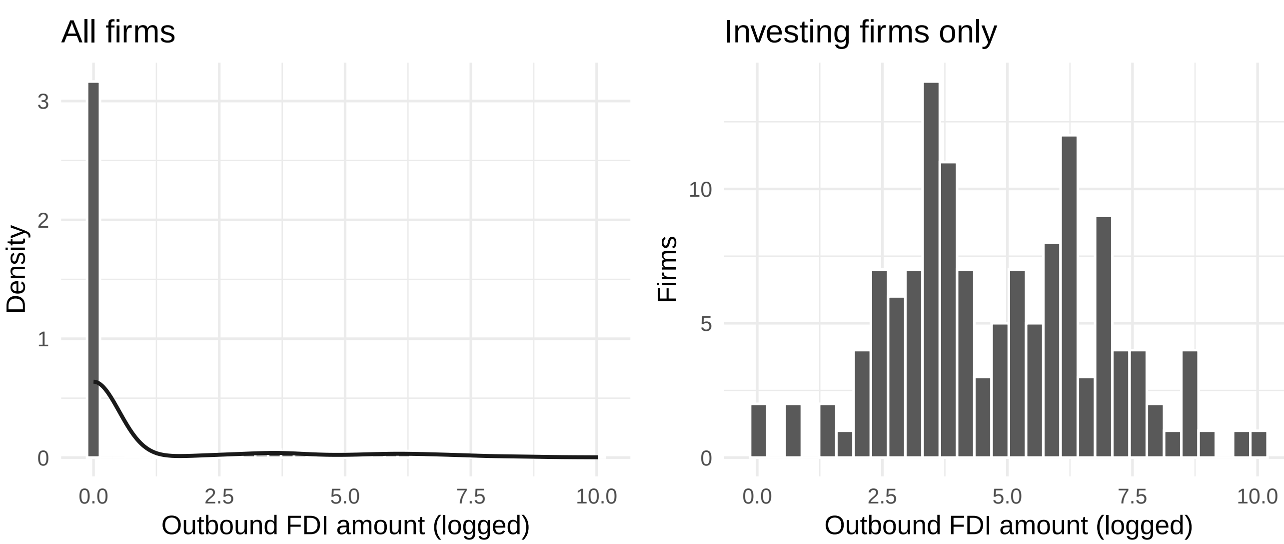

Figure 1: Left: the full outcome, dominated by a spike at zero (81% of firms never invest abroad). Right: conditional on investing at all, the positive amounts are right-skewed. Together these motivate the non-Gaussian residual diagnostics later.

Table 2: Distribution of the continuous outcome (logged outbound FDI amount).

Statistic	Value
N	703
Mean	0.916
SD	2.098
Min	0
Q1	0
Median	0
Q3	0
Max	10.027
IQR	0
Skewness	2.21
Kurtosis (excess)	3.746
% exactly zero	81.1%
Mean | positive	4.843

Shape, skew, and transformation. The outcome is strongly right-skewed (skewness 2.21) and leptokurtic (excess kurtosis 3.75), driven by the enormous floor at zero: the median, Q1, and Q3 are *all* zero, so more than three-quarters of firms sit exactly at the boundary. This is a **floor/zero-inflation** problem, not ordinary skew — the outcome is already a log of the raw amount, so a further monotone transform would not remove the spike (you cannot spread out a point mass at zero). A principled treatment would be a hurdle or Tobit model; the paper stays with OLS, and the price is paid in the residual diagnostics. For evaluation purposes the practical implication is that a naive mean-only predictor is hard to beat on RMSE, because the majority class is a single value.

2.3 Univariate distributions of the predictors

The study has one continuous headline confound (**firm size**), two continuous financial controls (**debt ratio**, **profitability**), two binary indicators (**board ties to state banks** — the treatment — and **chaebol** status), and a nine-level industry factor (**SIC category**). We profile the numeric predictors first (table + faceted histograms), then the categorical/binary ones as frequency tables and bars.

Table 3: Numeric predictors (raw data, before listwise deletion).

Predictor	n	n missing	Mean	SD	Min	Q1	Median	Q3	Max	Skew
Firm size (scaled mkt cap)	726	6	1.36	6.29	0.00	0.05	0.12	0.5	137.95	15.69
Debt ratio	728	4	0.26	0.19	0.00	0.10	0.24	0.4	0.84	0.50
Profitability (ROE)	709	23	-2.01	67.11	-1307.74	0.25	6.32	12.2	441.23	-12.23

The three numeric predictors are on wildly different scales and shapes, so the faceted histograms below use free x-axes. Firm size is extraordinarily right-skewed (a few giant firms dwarf the rest); debt ratio is a bounded proportion; and ROE has a small number of extreme outliers in both tails that stretch the axis.

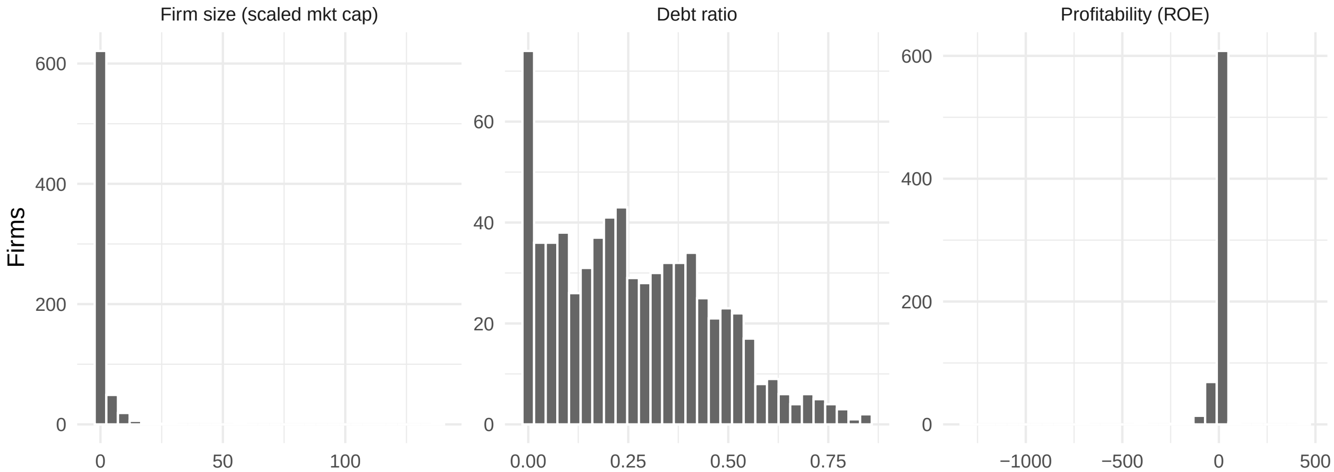

Figure 2: Univariate distributions of the three numeric predictors (free x-axes). Firm size is severely right-skewed; debt ratio is a bounded proportion; profitability (ROE) carries extreme two-tailed outliers.

Table 4: Binary predictors in the modeling frame (proportion equal to 1).

Binary predictor	= 0 (count)	= 1 (count)	Proportion = 1
Board ties to state banks	611	92	0.131
Chaebol (large conglomerate)	565	138	0.196

Both binary predictors are **imbalanced**: only about 13% of firms have board ties to state banks, and roughly 20% are chaebol-affiliated. The industry factor spreads firms across nine one-digit SIC categories, one of which dominates.

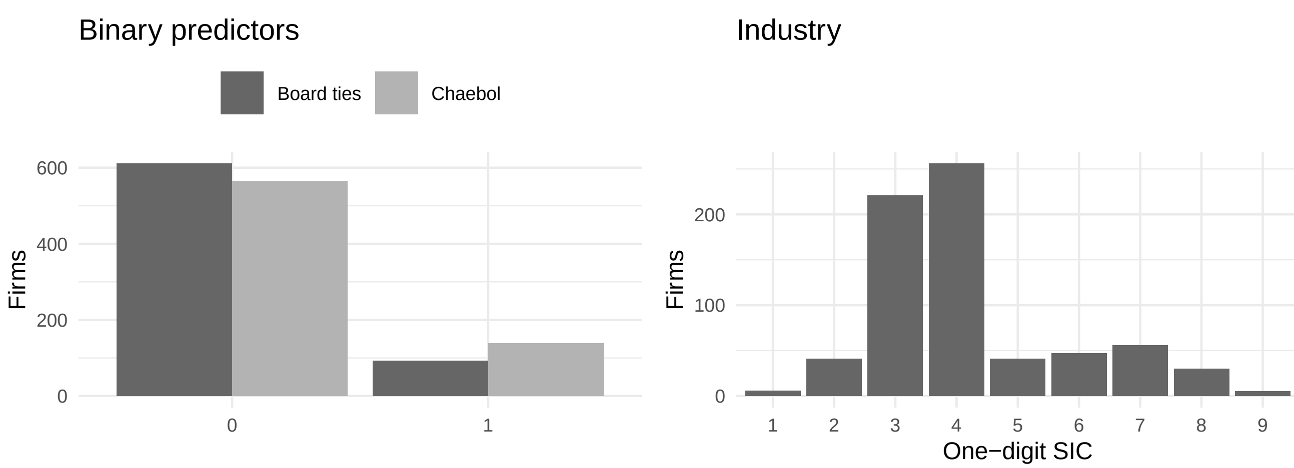

Figure 3: Frequency of the categorical predictors: the two binary indicators (left, both minority-1) and the nine-level SIC industry factor (right, one category dominant).

Table 5: Industry composition: nine one-digit SIC categories.

SIC category (one digit)	Firms	Share
1	6	0.9%
2	41	5.8%
3	221	31.4%
4	256	36.4%
5	41	5.8%
6	47	6.7%
7	56	8.0%
8	30	4.3%
9	5	0.7%

2.4 Missing-data analysis

Listwise deletion removes 29 of the 732 raw firms (4.0%). The table below counts and percentages missing values per model variable in the raw data.

Table 6: Missingness per model variable (of 732 raw firms).

Variable	# missing	% missing
Profitability	23	3.1%
SIC category	8	1.1%
Firm size	6	0.8%
Debt ratio	4	0.5%
Chaebol	1	0.1%
Outbound FDI amount	0	0.0%
Board ties to state banks	0	0.0%

Table 7: Missingness pattern: complete vs. incomplete firms and which variables drive the loss.

Quantity	Firms
Complete cases (all 7 vars present)	703
Incomplete firms (>= 1 var missing)	29
… missing Firm_Size	6
… missing Debt_Ratio	4
… missing Profitability	23
… missing Chaebol	1
… missing sic_category	8

The loss is concentrated in a couple of variables rather than scattered evenly, and no single firm is missing many fields at once — the incomplete rows are mostly missing a single value. There is no obvious sign that missingness is tied to the outcome (the outcome itself is essentially complete), so listwise deletion of ~4% of firms is unlikely to bias the headline estimates much; the reproduction handles it by the same `complete.cases()` listwise deletion the authors used.

2.5 Bivariate relationships with the outcome

We now relate the outcome to each key predictor. For the numeric predictors we scatter FDI against the predictor with a LOESS smoother and report the Pearson correlation; for the categorical/binary predictors we compare group means. Firm size is the dominant continuous driver and the treatment (board ties) is the substantive quantity of interest, so those two get the most attention.

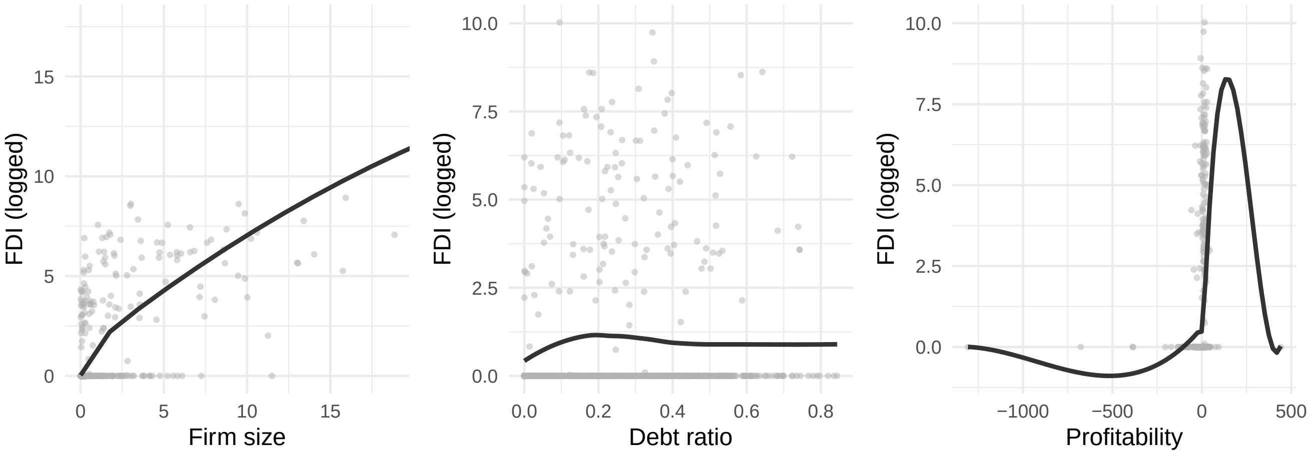

Figure 4: Outbound FDI vs. each numeric predictor with a LOESS smoother (firm size on a 1-99 percentile window). Firm size rises steeply and then bends; debt ratio and profitability are essentially flat against the outcome.

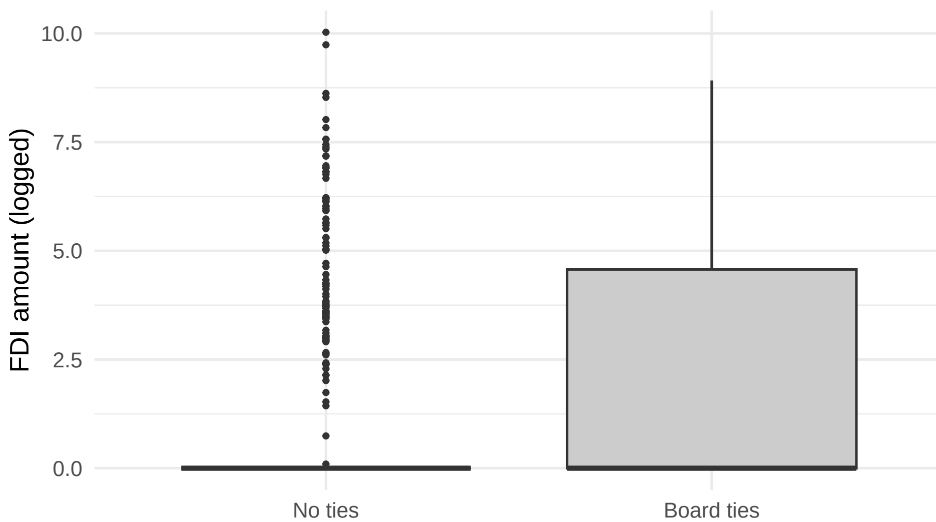

Figure 5: FDI amount by board-tie status; tied firms sit higher on average, the paper's headline raw association.

Table 8: Marginal association of each predictor with the outcome (correlation for numeric, group-mean gap / eta-squared for categorical).

Predictor	Type	Assoc. with FDI
Firm size	numeric	r = 0.472
Debt ratio	numeric	r = 0.037
Profitability	numeric	r = 0.078
Board ties	binary	mean diff = 1.201

Predictor	Type	Assoc. with FDI
Chaebol	binary	mean diff = 1.655
SIC industry	factor(9)	eta^2 = 0.012

Table 9: Outcome by board-tie status: tied firms invest abroad more often and by more.

Board ties	Firms	Mean FDI	% investing abroad
No ties	611	0.759	16%
Board ties	92	1.960	36%

Firm size carries essentially all of the marginal signal (r = 0.47 with the outcome), and the scatter *curves* rather than rising in a straight line. Board ties show a clear positive raw gap, but as the next subsection shows, tied firms also tend to be larger — a confound we expect the regression to partial out.

2.6 Multivariate structure and collinearity

Finally we look at the joint structure among the numeric variables — a correlation heatmap, a compact pairs panel, and the variance-inflation factors from the reproduction model — to check whether any predictors are dangerously collinear.

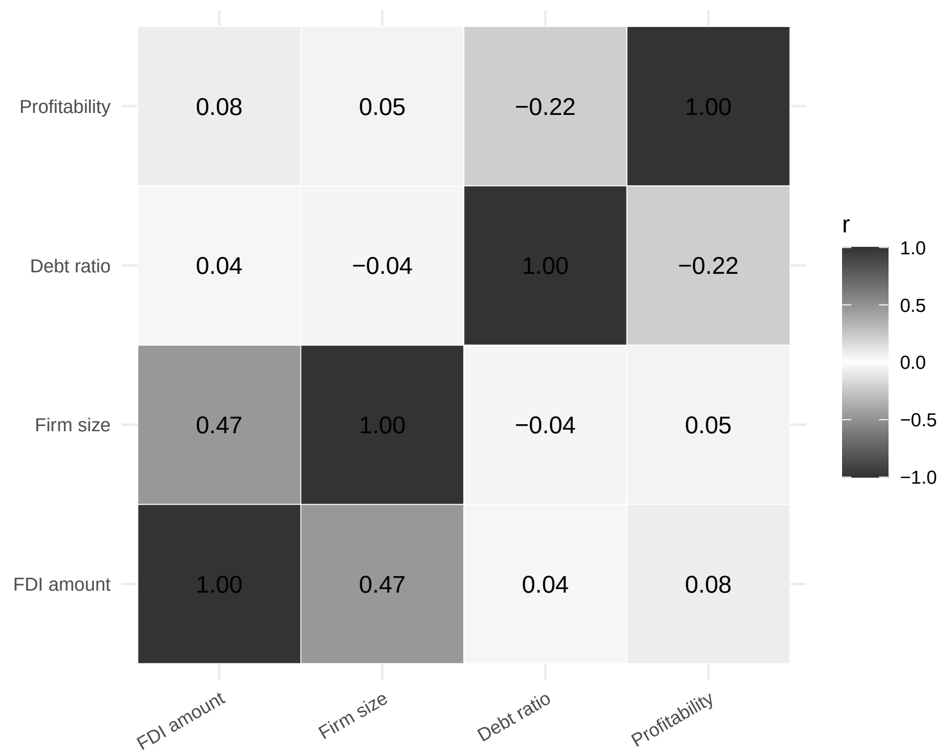

Figure 6: Correlation heatmap among the numeric variables (outcome + three continuous predictors). The only strong cell is firm size with the FDI outcome; the continuous controls are near-orthogonal to one another.

Table 10: Pearson correlations among the numeric variables (outcome + continuous predictors).

	FDI amount	Firm size	Debt ratio	Profitability
FDI amount	1.000	0.472	0.037	0.078
Firm size	0.472	1.000	-0.044	0.051
Debt ratio	0.037	-0.044	1.000	-0.220
Profitability	0.078	0.051	-0.220	1.000

Table 11: Variance-inflation factors from the reproduction model (SIC dummies excluded from display). All near 1 – no collinearity concern.

Term	VIF
Board ties to state banks	1.30
Firm size	1.08
Debt ratio	1.11
Profitability	1.10
Chaebol	1.18

The heatmap has a single strong cell (firm size with the FDI outcome, r = 0.47); every predictor-with-predictor correlation is small. The VIFs are all close to 1 — far below any rule-of-thumb threshold (5 or 10) — so there is **no collinearity** worth worrying about (no $|r| > 0.7$ anywhere). That is good news for interpretation: each slope in the reproduction is cleanly identified, and the modeling can focus on *functional form* rather than variance inflation.

What the descriptives foreshadow. The outcome is dominated by a spike at zero and a skewed positive tail, which is exactly why the OLS residual diagnostics later look non-Gaussian and heteroscedastic. Among the predictors, only firm size is strongly correlated with FDI, and that relationship visibly *curves* — motivating the polynomial, spline, and GAM extensions that are the heart of this Day-2 tutorial. Board ties, the treatment, show a clear positive raw association with FDI, but tied firms are also larger, so we should expect firm size to absorb part of that association once both enter the regression together. Collinearity is not a worry (VIFs ≈ 1), so the modeling can focus squarely on functional form rather than variance inflation.

3 Reproduction of the published regression

We re-estimate Table 1, models 1–4, on the downloaded data. The values labelled "Published" are the numbers produced by the authors' own replication script (which by construction equal the printed table); "Reproduced" are our re-run of the identical `lm()` specification. They match to reported precision.

Table 12: Table 1, model 4: published vs. reproduced OLS coefficients.

Term	Published (m4)	Reproduced (m4)
Board ties to state banks	0.548	0.548
Firm size	0.136	0.136
Debt ratio	0.542	0.542
Profitability	0.002	0.002
Chaebol (large conglomerate)	1.023	1.023

Table 13: Reproduced Table 1 across specifications (key terms).

Model	Board ties	Firm size	N	R2
m1: ties only	1.240	NA	724	0.043
m2: size only	NA	0.155	720	0.229
m3: ties + size	0.789	0.150	720	0.242
m4: full	0.548	0.136	703	0.281

The board-ties coefficient is positive and significant throughout ($\hat{\beta} = 0.548$, $p = 0.017$ in the full model), and firm size is the dominant predictor ($\hat{\beta} = 0.136$). **The reproduction is exact.** Everything below extends this same model without disturbing it.

4 OLS setup and multiple-regression interpretation

The reproduced model 4 is an ordinary least squares fit $y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$, where $\hat{\beta}$ minimizes the residual sum of squares $\sum_i (y_i - \hat{y}_i)^2$ and, under the classical assumptions, is BLUE. In a *multiple* regression each slope is a **partial** effect: the expected change in the (logged) FDI amount for a one-unit change in that predictor, *holding all the others fixed.* The coefficient table below reads off model 4 with this interpretation in mind.

Table 14: Model 4 OLS coefficients (SIC dummies estimated but omitted from display).

Term	Estimate	Std. error	t value	p value
Board ties to state banks	0.548	0.228	2.40	0.017
Firm size	0.136	0.011	12.33	0.000
Debt ratio	0.542	0.364	1.49	0.137
Profitability	0.002	0.001	1.63	0.105
Chaebol	1.023	0.185	5.54	0.000

- **Board ties to state banks** (+0.548): a firm whose board is tied to a state bank has, on average, a logged FDI amount 0.55 higher than an otherwise-identical firm without such ties — the paper's headline result.
- **Firm size** (+0.136): the dominant term. Each one-unit (one-million-USD) increase in the market-cap measure is associated with a 0.136 rise in logged FDI, controlling for the rest. We will show below this straight-line slope hides a strong curvature.
- The remaining controls (debt ratio, profitability, chaebol status, and nine SIC-industry dummies) are partialled out of every coefficient above.

Overall the model explains $R^2 = 0.281$ of in-sample variance (adjusted $R^2 = 0.268$), on N = 703.

5 Residual diagnostics

The four standard `plot.lm` diagnostics probe the OLS assumptions. The figure below shows all four for model 4.

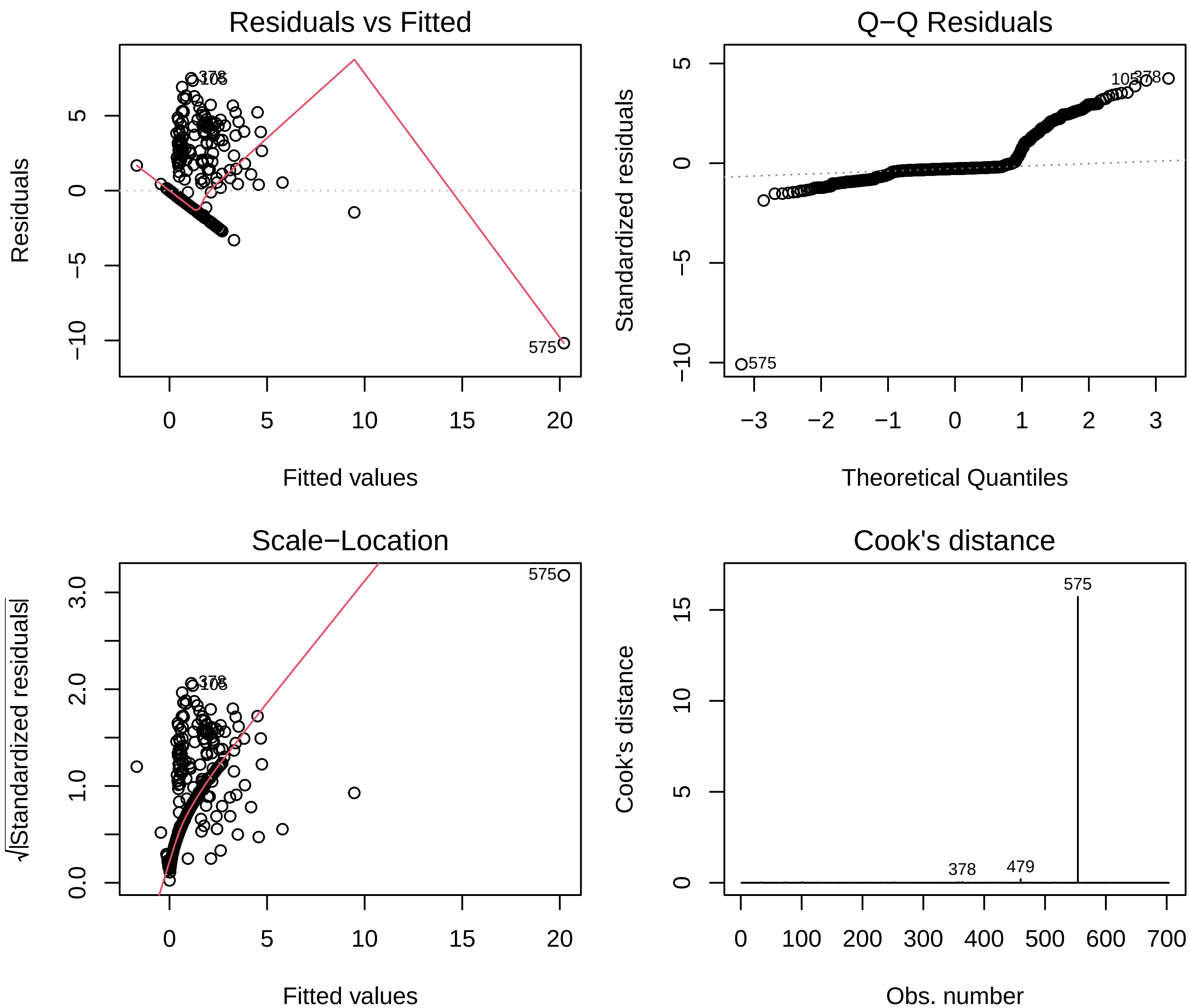

Figure 7: Residual diagnostics for the reproduced OLS model 4: (top-left) residuals vs fitted for linearity/mean-zero errors; (top-right) normal QQ for the normality of errors; (bottom-left) scale-location for homoscedasticity; (bottom-right) residuals vs leverage with Cook's-distance contours for influential points.

Reading the panels:

- **Residuals vs fitted** — the smoother is not flat and the point cloud fans out with the fitted value: the conditional mean is *not* well captured by the linear terms, and the spread grows. Both are symptoms of the outcome's heavy zero-mass and skew.
- **Normal QQ** — the upper tail departs sharply from the reference line; errors are far from Gaussian (again the FDI distribution: 81% of firms have exactly zero).
- **Scale-location** — the rising $\sqrt{|\text{std. resid}|}$ trend confirms **heteroscedasticity**, so classical OLS standard errors are suspect.
- **Residuals vs leverage / Cook's distance** — a handful of high-leverage, high-residual firms sit near or beyond the Cook's-distance contours; the most influential observations are flagged below.

Table 15: Five most influential firms by Cook's distance (rule-of-thumb cutoff 4/n = 0.0057).

Row	Cook's D	Leverage	Std. resid
554	15.719	0.684	-10.09
460	0.203	0.578	1.44
364	0.043	0.032	4.25
357	0.037	0.047	3.24
101	0.036	0.029	4.15

Because heteroscedasticity is evident, the table below contrasts the classical and heteroscedasticity-consistent (HC3, White) standard errors for the two headline terms. The point estimates are unchanged (they must be — same OLS fit), but honest inference should use the robust column.

Table 16: Classical vs. heteroscedasticity-consistent (HC3) standard errors.

Term	Estimate	Classical SE	Robust SE (HC3)
Board ties to state banks	0.548	0.228	0.565
Firm size	0.136	0.011	0.163

6 Logistic regression and MLE (pedagogical aside)

The published model is a continuous OLS; this section is a clearly-labeled ***pedagogical aside*** *to illustrate logistic regression and maximum likelihood — it is not part of the paper's design.* We binarize the outcome into **any outbound FDI** — since 81% of firms have exactly zero FDI, `FDI > 0` is the natural cut (a "did the firm invest abroad at all?" event, 19% positive) — and fit a logistic GLM with the same predictors.

Where OLS minimizes squared error, logistic regression maximizes the **Bernoulli log-likelihood** $\ell(\beta) = \sum_i \{y_i \log p_i + (1 - y_i)\log(1 - p_i)\}$ with $p_i = \text{logit}^{-1}(x_i'\beta)$. Coefficients are **log-odds**: $\hat{\beta}_j$ is the change in the log-odds of any-FDI per unit of x_j; $e^{\hat{\beta}_j}$ is the odds ratio. The model is fit by IRLS; the printed **deviance** is -2ℓ at the solution, and the difference between null and residual deviance is the likelihood-ratio evidence that the predictors matter.

Table 17: Logistic regression for Pr(any outbound FDI). Coefficients are log-odds; the odds-ratio column exponentiates them.

Term	Log-odds (beta)	Odds ratio	Std. error	p value
Board ties to state banks	-0.722	0.486	0.435	0.097
Firm size	0.709	2.031	0.092	0.000
Debt ratio	1.373	3.946	0.651	0.035
Profitability	0.005	1.005	0.004	0.144
Chaebol	0.782	2.186	0.292	0.007

Table 18: Maximum-likelihood fit statistics for the logistic model.

Quantity	Value
Log-likelihood at MLE	-232.6
Null deviance	682

Quantity	Value
Residual deviance	465.2
LR chi-square (null - residual)	216.7 on 13 df (p <2e-16)
In-sample AUC	0.84

Reading the aside carefully is itself instructive. **Firm size** dominates the log-odds of investing abroad at all (a large, highly significant positive coefficient), exactly as in the continuous model. **Board ties**, however, illustrate a classic partial-vs-marginal subtlety: *unconditionally*, tied firms invest abroad far more often (36% vs 16%), but once firm size and industry are held fixed the *conditional* log-odds coefficient is small and not significant (odds ratio = 0.49, $p = 0.097$) — tied firms tend to be large firms, and size absorbs the raw association. This is a reminder that binarizing a continuous outcome discards information and can shift a coefficient's apparent role; the paper's finding lives on the *amount* of FDI, not the yes/no margin. The likelihood-ratio test nonetheless overwhelmingly rejects the null of no predictors, driven by firm size.

7 Flexible mean functions in firm size

Firm size is the dominant predictor, and the diagnostics hinted its effect is not a straight line. We now model the FDI–size relationship with progressively more flexible mean functions, holding the other model-4 controls linear throughout so the comparison isolates the *functional form of size*:

1. **Linear** — the model-4 term, `Firm_Size` entered linearly.
2. **Polynomial** — `poly(Firm_Size, 3)`, a cubic.
3. **Natural spline** — `splines::ns(Firm_Size, df = 5)`.
4. **B-spline** — `splines::bs(Firm_Size, df = 6)`.
5. **GAM** — `mgcv::gam` with a penalized smooth `s(Firm_Size)`; the smoothness is chosen automatically by GCV/REML rather than fixed by hand.

7.1 Polynomial vs linear

A partial-F test of the cubic against the linear term asks whether the extra curvature buys significant in-sample fit.

Table 19: Nested-F test: cubic polynomial in firm size vs. the linear term (other controls identical).

Comparison	Resid. df	RSS	F	p value
Linear	689	2219.7	NA	NA
Cubic polynomial	687	1569.0	142.45	<2e-16

7.2 Fitted size curves

To visualize each fit's implied FDI–size curve we predict over a grid of firm size while holding every other predictor at a representative value (median for continuous, modal SIC, no board ties, non-chaebol). The figure below overlays them on the raw data.

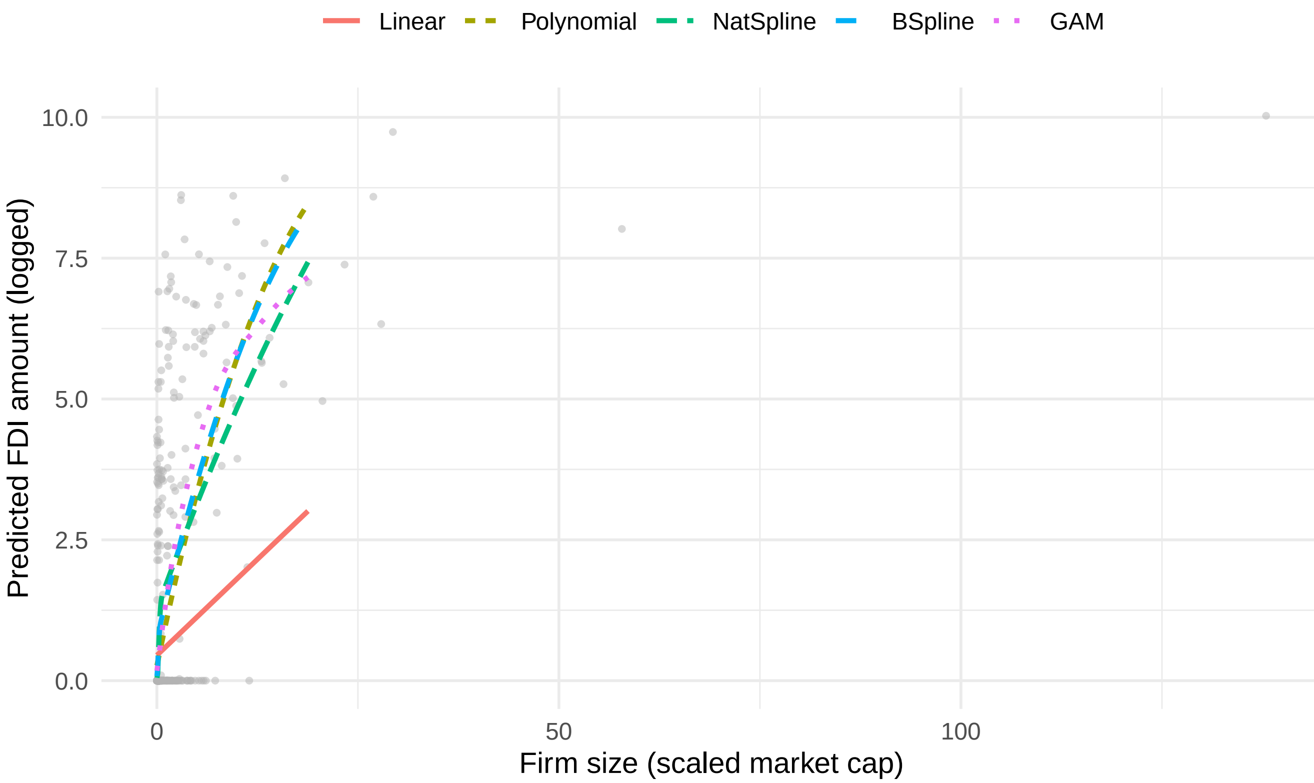

Figure 8: Fitted FDI-vs-firm-size curves from five mean functions, holding other predictors at representative values. The linear term is monotone and misses the rise at large firm size; the spline and GAM track a steeper, saturating nonlinearity.

7.3 The GAM smooth

The figure below plots the penalized smooth $s($`Firm_Size`$)$ from `mgcv::gam` directly (partial effect, other terms held at their means), with the estimated degrees of freedom that quantify how far it bends away from a straight line.

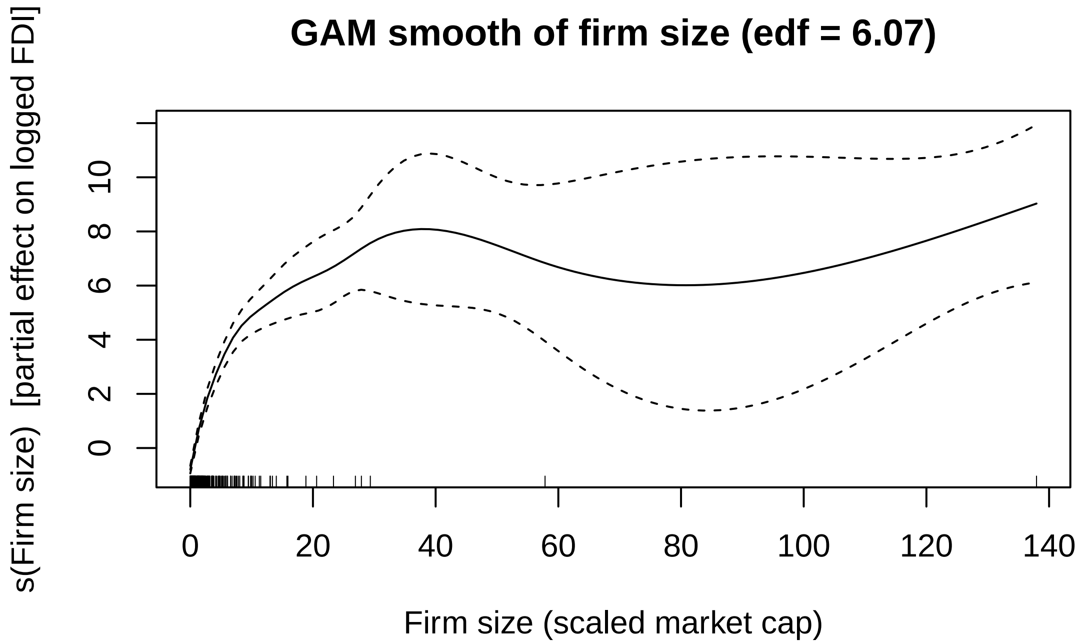

Figure 9: GAM penalized smooth of firm size on logged FDI (mgcv, REML). The shaded band is the pointwise 95% CI; the effective degrees of freedom exceed 1, confirming statistically detectable curvature.

The smooth's effective degrees of freedom is 6.07 (a straight line would be exactly 1), and the smooth term is significant ($p < 2e - 16$): the FDI–size relationship genuinely curves.

8 Out-of-sample comparison of all extensions

In-sample fit always improves with flexibility, so the honest test is held-out prediction. Under a common 10-fold cross-validation we score each mean function by out-of-sample RMSE and R^2 on the continuous FDI outcome. All five share the identical control set; they differ only in how firm size enters.

Table 20: Out-of-sample prediction (10-fold CV), continuous FDI outcome. All models share the model-4 controls; only the firm-size term differs. Naive mean-only baseline RMSE = 2.096. Lower RMSE / higher R2 is better.

Mean function	CV RMSE	CV R2
Linear	2.125	-0.028
Polynomial	11.695	-30.133
NatSpline	1.668	0.367
BSpline	9.922	-21.407
GAM	1.600	0.417

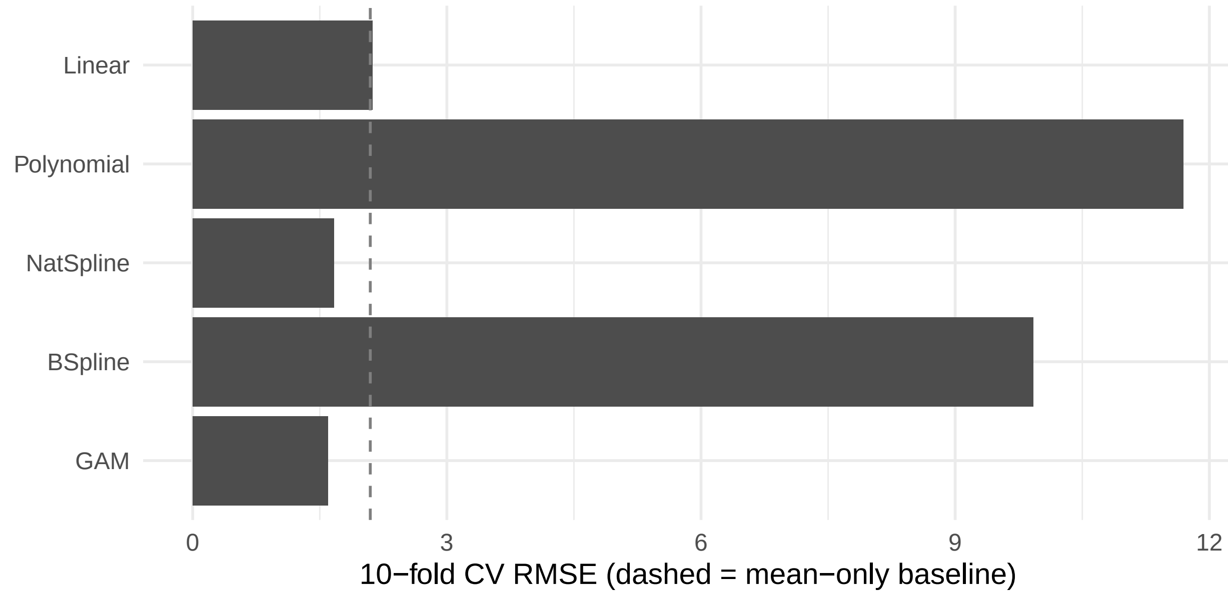

Figure 10: Cross-validated RMSE by mean function. Moving from a straight line to a natural spline or GAM in firm size lowers held-out error. The cubic polynomial and high-df B-spline collapse out of sample (large negative CV R-squared): global or unpenalized bases fitted on a training fold extrapolate wildly at test points near and beyond that fold's firm-size range. Note also that the linear-in-size model trails even the mean-only baseline.

9 Takeaway

The published OLS result reproduces **exactly**: board ties to state banks raise firm FDI ($\hat{\beta} = 0.548$, $p = 0.017$), with firm size the dominant driver ($\hat{\beta} = 0.136$). Reading that model as a proper multiple regression, each slope is a partial effect; but its residual diagnostics are unambiguous — non-flat residuals-vs-fitted, a heavy-tailed QQ plot, clear heteroscedasticity (so we report HC3 robust SEs), and a few influential firms. The logistic / MLE aside re-expresses the analysis on a binary "any-FDI" outcome via the log-likelihood, deviance, and log-odds: firm size again dominates, while board ties show a positive *marginal* association that shrinks to insignificance once size is controlled — a reminder that binarizing a continuous outcome can move a coefficient's apparent role, and that the paper's result lives on the FDI amount.

The core Day-2 lesson is about **functional form**. A partial-F test rejects the linear firm-size term in favour of a cubic; a `mgcv::gam` smooth has effective df well above 1 and is significant; and the natural-spline and GAM curves in the fitted-curve figure bend where the straight line cannot. Crucially, cross-validation keeps this honest: the natural spline and GAM lower held-out RMSE relative to the linear term, whereas the unpenalized cubic polynomial and high-df B-spline fail catastrophically out of sample (CV R^2 below -20) — an extrapolation failure of global, unpenalized bases at the edges of the firm-size range, and the clearest argument in this tutorial for local spline bases and penalized smooths. The linear-in-size model even trails the mean-only baseline out of sample (CV $R^2 = -0.03$), so getting the functional form right is not cosmetic here. Flexible regression sharpens the firm-size relationship's shape while leaving the sign and substance of the paper's argument intact.

10 Independent exercises (each about 10 minutes)

1. **A second smooth.** Refit the full-data GAM replacing `Debt_Ratio` with `s(Debt_Ratio)` (keep `method = "REML"`). From `summary()`: what is the effective df of the new smooth, and is it distinguishable from a straight line?

2. **Does it generalize?** Swap the two-smooth formula into the CV loop's GAM line and re-run the `cv` chunk. Does held-out RMSE improve on the reported GAM?
3. **Residual plots as a functional-form check.** The slides stress residual plots as the first diagnostic for a misspecified mean function. Plot residuals against fitted values for the linear fit and for the GAM (`plot(fitted(fit_lin), resid(fit_lin))`; same for `fit_gam`). Does the curvature visible in the linear model's residuals disappear once firm size enters as a smooth?